

How to Compare Nested Models in R

Posted on May 9, 2021 by George Pipis in R bloggers | 0 Comments

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Using R and the `anova` function we can easily compare **nested** models. Where we are dealing with regression models, then we apply the `F-Test` and where we are dealing with logistic regression models, then we apply the `Chi-Square Test`. By nested, we mean that the independent variables of the **simple** model will be a **subset** of the more **complex** model. In essence, we try to find the best parsimonious fit of the data. Note that we should fit the models on the same dataset.

The Null Hypothesis is that the **simple** model is better and we reject the null hypothesis if the p-value is less than 5% inferring that the **complex** model is significantly better than the simple one.

Example of Comparing Nested Models

Let's work with the `LifeCycleSavings` dataset by considering as dependent variable the `sr` and the rest as independent variables (IV).

> `LifeCycleSavings`

	sr	pop15	pop75	dpi	ddpi
Australia	11.43	29.35	2.87	2329.68	2.87
Austria	12.07	23.32	4.41	1507.99	3.93
Belgium	13.17	23.80	4.43	2108.47	3.82
Bolivia	5.75	41.89	1.67	189.13	0.22
Brazil	12.88	42.19	0.83	728.47	4.56
Canada	8.79	31.72	2.85	2982.88	2.43
Chile	0.60	39.74	1.34	662.86	2.67
China	11.90	44.75	0.67	289.52	6.51
Colombia	4.98	46.64	1.06	276.65	3.08
Costa Rica	10.78	47.64	1.14	471.24	2.80
Denmark	16.85	24.42	3.93	2496.53	3.99
Ecuador	3.59	46.31	1.19	287.77	2.19
Finland	11.24	27.84	2.37	1681.25	4.32
France	12.64	25.06	4.70	2213.82	4.52
Germany	12.55	23.31	3.35	2457.12	3.44
Greece	10.67	25.62	3.10	870.85	6.28
Guatemala	3.01	46.05	0.87	289.71	1.48
Honduras	7.70	47.32	0.58	232.44	3.19

Let's say that we can to compare the following two models:

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- `fit0` which is the $sr = \alpha$ VS
- `fit1` which is the $sr = \alpha + \beta \times pop15$

```
fit0 <- lm(sr ~ 1, data = LifeCycleSavings)

fit1 <- lm(sr ~ pop15, data = LifeCycleSavings)

summary(fit0)
summary(fit1)
```

```
> summary(fit0)
```

```
Call:
lm(formula = sr ~ 1, data = LifeCycleSavings)

Residuals:
    Min       1Q   Median       3Q      Max
-9.071 -2.701  0.839  2.946 11.429

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  9.6710     0.6336   15.26  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.48 on 49 degrees of freedom
```

```
> summary(fit1)
```

```
Call:
lm(formula = sr ~ pop15, data = LifeCycleSavings)

Residuals:
    Min       1Q   Median       3Q      Max
-8.637 -2.374  0.349  2.022 11.155

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 17.49660     2.27972   7.675 6.85e-10 ***
pop15       -0.22302     0.06291  -3.545 0.000887 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.03 on 48 degrees of freedom
Multiple R-squared:  0.2075,    Adjusted R-squared:  0.191
F-statistic: 12.57 on 1 and 48 DF, p-value: 0.0008866
```

Notice the the P-value of the F-Test of the `fit1` model is **0.0008866** which actually actually tests the Null Hypothesis that “all the beta coefficients are zero” versus the alternative hypothesis that “at least one beta coefficient is not zero”. Since we have only one beta coefficient, the `pop15` the p-value of the F-Test is the same with the p-value of the T-Test as we can see above.

Now, if we compare the `fit0` vs the `fit1`, in essence, we test if we should include the `pop15` coefficient or not, thus we expect to get the same p-value. Let's compare the nested models using `anova`:

```
anova(fit0, fit1, test='F')
```

As, expected we got the same p-value, and we can say that we should prefer the `fit1` compared to `fit0` model.

Let's make another comparison by comparing the `fit1` compared to the `fit4` which contains all the IVs.

```
fit4<-lm(sr~pop15+pop75+dpi+ddpi, data = LifeCycleSavings)

summary(fit4)
```

Let's compare the two models:



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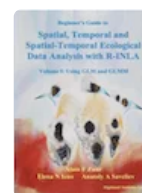


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```
anova(fit1, fit4, test='F')
```

The p-value is **0.04177** forcing us to reject the null hypothesis that the `fit1` models is better. Finally, let's compare the `fit1` model versus the `fit3` which contains the first 3 IV of the dataset.

```
fit3<-lm(sr~pop15+pop75+dpi, data = LifeCycleSavings)
anova(fit1, fit3, test='F')
```

In this case, the p-value is **0.1318** which means that we should accept the null hypothesis that the `fit1` is better than the `fit3`.

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